

PART - A

- Q. No. 1. [b] Answer: A dedicated application.
2. [b] Answer: Data processing and control.
3. [c] Answer: Real-Time operating system.
4. [a] Answer: Power Management Unit.
5. [c] Answer: AT Mega 328P.
6. [c] Answer: loop().
7. [b] Answer: To simulate varying voltages between 0V and 5V.
8. [c] Answer: Arduino Mega 2560.
9. [b] Answer: DHT11 sensor.
10. [c] Answer: HC-SR04 ultrasonic.

SECTION - B

21. b) Embedded Systems:

Definition:

An embedded system is a system which has embedded software and computer hardware working together to complete a specific functions, applications, etc.

Embedded Software:

An embedded software is the software where the system codes are stored. The software programs are found in embedded software.

Computer Hardware:

A computer hardware is which external devices are connected. For example,

i) Microprocessor.

ii) Microcontroller

iii) I/O devices, etc.

CHARACTERISTICS:

i) Embedded device:- A device with embedded software must be present.

ii) Low-power consumption:

Since most of the devices are battery powered, low power consumption is required.

iii) Memory:

It has ~~no~~ low-memory. only less data will be stored.

iv) User Interaction:

User can not alter any embedded device because it is present in a larger system.

CONSTRAINTS:

i) Low - Power Consumption / Power Dissipation:

The power consumption is limited or low because most of the devices are battery powered.

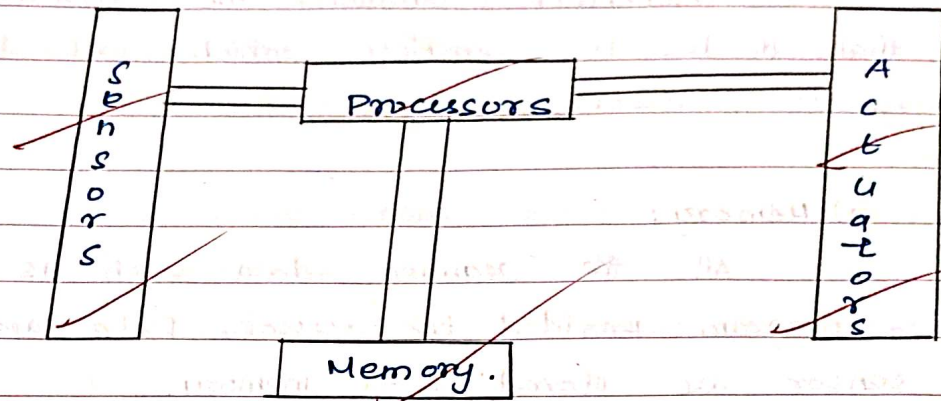
ii) Speed of Processor:

The device will work according to the certain level that a processor can complete. It will not work well when exceeded.

iii) Memory consumption:

Memory consumption is low because, it has only limited memory space. So it contains specific information/tasks of the device.

DIAGRAM:



WORKING:

i) Processors :

The processors contain the informations of the input data & the functions. It receives data from the sensors and provides the makes decision based on the information provided and it passes what work has to be completed to the actuators.

ii) Sensors :

Sensors are the input source. Sensors get information from the surrounding or from required source and provides the information to the processor. This data is used by the processor.

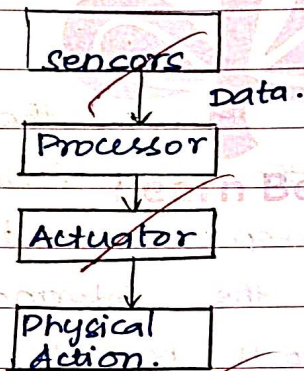
iii) Actuators:

Actuators completes the physical actions that it has to complete which was instructed by the processor.

iv) Memory:

All the required data such as ~~pro~~ information & program provided to processor, Data read by the sensor are stored in the memory.

WORKFLOW:



CONCLUSION:

Hence in smart greenhouse systems, the DHT11/DHT22 sensors checks the temperature & humidity levels and it sends the data to processors, then the processor decides what to do and sends it to the actuators and it does physical action.

22. a) ARDUINO PROGRAMMING:

(i) setup():

- setup() is the function used to initialize.
- It is a function where initialization occurs & program starts.
- It runs only once and moves to the next.

Example:

```
void setup()  
{  
    pinMode(13, OUTPUT);  
    serial.begin(9600);  
}
```

ii) loop():

- loop() is the function where the implementation runs.
- It runs continuously.
- The block of code given here is run, the code is implemented here.

Example:

```
void loop()  
{  
  pinMode(13, HIGH);  
  delay(1000);  
}
```

CONDITIONAL STATEMENTS :

i) if :

The if statement will run if the condition given is true / satisfied.

Eg: if (a > 0):
c = a + b;

ii) if - else :

If the if statement doesn't satisfy then the else statement will run.

Eg: if (a > 0):
c = a + b;
else:
print("0");

iii) else - if:

When there are many if conditions else-if is used.

DIGITAL I/O statements:

i) PinMode :

Gets input to the pin.

ii) DigitalWrites :

will write the input to the interface.

iii) DigitalRead :

Will get the input from the user or interface.

delay() :

It will take time to complete the operation.

CONCLUSIONS :

Using these only the Traffic signal controllers repeatedly execute predefined sequences while responding to changing traffic conditions.

Q3.a) WATER-LEVEL SENSOR:

AIM:

A residential building requires an automated security system to monitor the water level in an overhead tank and control the pump. It has to be implemented.

REQUIREMENTS:

- i) water-level sensor - To measure the level of water.
- ii) HIGH, LOW water sensors - For high/low level water.
- iii) Water pump.
- iv) Relay Module
- v) Arduino Uno.

WORKING:

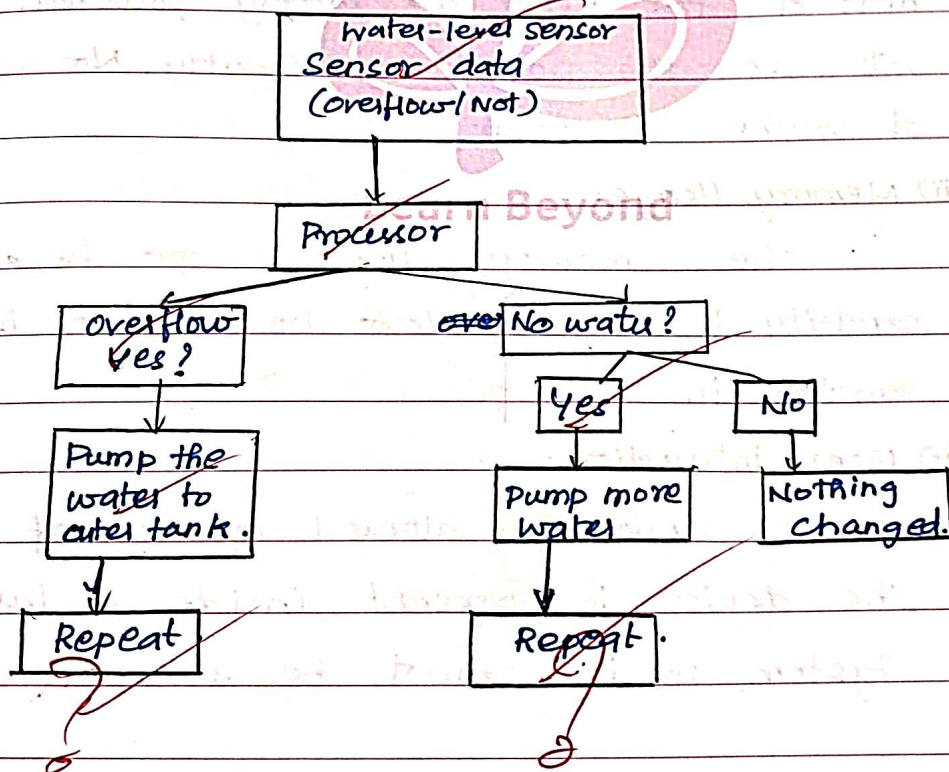
1. The level of water is first determined by the water level sensor.
2. When there is HIGH water level the HIGH-water sensor will send information to the processor. & if it is low then LOW-water sensor will send.
3. When something is noted the processor will make decision whether the

water has to be pumped or pumped out to the cante tank.

4. Then after making decision the processor will tell it to actuator and it will do the physical action.

The main mechanism used here is if-else conditional statements.

WORKFLOW:



SECTION - B.

II. CHARACTERISTICS OF AN EMBEDDED SYSTEM:

i) Specific Task:

It is mainly used to complete a specific task. It doesn't do multitasking.

Eg: Washing Machine - only washes clothes.

ii) Power consumption:

Power consumption is relatively low because most of the devices are battery powered. So it will use the power efficiently. No wastage of power will occur.

iii) Memory Usage:

The memory used is low because its capacity is already low because it is very smaller in size.

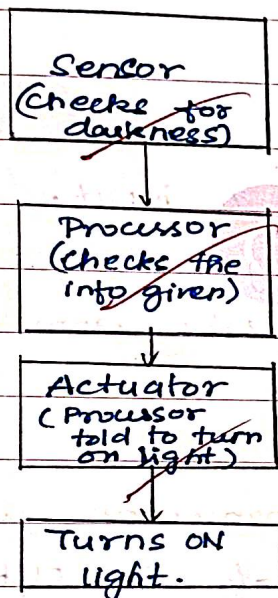
iv) User interaction:

It cannot be altered by any of the users. The device is present inside a larger system, so it cannot be altered.

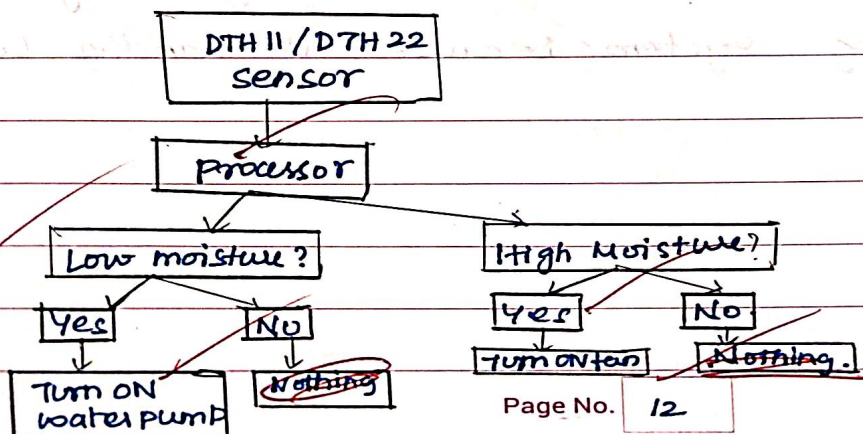
12. Interrupt Service Routine (ISR):

The external interrupts are received by the ~~INT0~~ ~~&~~ ~~INT1~~ and these interrupts will be sent to the processor. The main role of the ISR is to receive the interrupts and send it to the processor.

13.



14.



15. Arduino Boards:

- i) Arduino Uno - General purpose board.
- ii) Arduino Nano - More complexed than Uno.
- iii) Arduino Leonardo - USB pin purposes.
- iv) Arduino Ethernet - Used for Ethernet purposes.
- v) Arduino Mega 2560 - Used for complexed functions.

16. Analog I/O function:

- i) pinMode - Reads the pin as input.
- ii) AnalogWrite - Writes in analog pin.
- iii) AnalogRead - Reads/gets data from PWD pins.

17. Arduino Mega 2560:

It uses ATmega 2560 processor with 16 MHz clock speed with a SRAM of 8 KB & EEPROM of 16 KB. So it can be used effectively for a smart parking system because it handles large data.

18. Arduino Function:

The Arduino function used to perform a repeated pick and place operations in a robotic arm is Arduino Mega 2560.

17. Temperature sensor: (DTH11/DTH22):

In a green-house monitoring system, the temperature level of the surroundings is very important. Here the temperature sensor will get input of the temperatures ~~using~~ from the place & will give it to the processor.

20. Ultrasonic sensor:

An ultrasonic sensor (HC-SR04) will send rays towards an object and the echos will be formed. Once it reaches the object it will reach back to the sensor. It calculates the time & distance using this. So this is how it will be used in water level monitoring system.